

# Gauri Prabhakar

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## Education

### Northeastern University

Master of Science in Artificial Intelligence

Boston, MA

Expected June 2027

### Chandigarh University

Bachelor of Engineering in Computer Science (AI and ML)

Chandigarh, India

June 2018 – May 2022

## Technical Skills

**Languages:** Python, C++, Rust, C#, Java

**AI/ML:** PyTorch, TensorFlow, Hugging Face, Ray RLlib, scikit-learn, NLP, Reinforcement Learning

**Frameworks & Databases:** FastAPI, Flask, Streamlit, .NET, MongoDB, PostgreSQL, Redis, Pinecone, MySQL

**Tools & Infrastructure:** Git, Docker, Kubernetes, Azure, CI/CD, RabbitMQ, APIGEE, Agile/Scrum

## Experience

### Machine Learning Engineer

Oct 2024 – June 2025

AYU Pune, India

- Led design of conversational AI covering prompt engineering, custom NLP pipelines and sentiment analysis, boosting session retention by **30%**.
- Integrated **OpenAI GPT** and **Hugging Face** models; reduced latency by **40%** through async **FastAPI**, caching, batching and prompt optimization; improved engagement by **25%**.

### Software Engineer

Sep 2022 – Sep 2024

SLB (Schlumberger) Pune, India

- Built **.NET** API endpoint with custom query parser mapping requests to **MongoDB** aggregation pipelines, reducing response times by **80%**; mentored six engineers across **80+** APIs.
- Wrote shared **APIGEE** flow policy fixing broken href links in API metadata, unblocking **50+** downstream consumers whose callback scripts depended on those links.
- Executed zero-downtime migration of **327,000+** hierarchical **MongoDB** records across full schema redesign and platform restructure with zero data loss.
- Scripted DB correction pipelines resolving daily inconsistencies causing message pile-ups and operational slowdowns on the rig.
- Independently identified a gap, built and pitched a RAG-style AI chatbot to leadership using internal docs, wiki pages and meeting transcripts as corpus.

### Machine Learning Engineer

Feb 2022 – Aug 2022

iSchoolConnect Remote

- Shipped internal **Streamlit** tool giving non-tech teams direct access to recommendation system, cutting cross-team dependency by **85%** and contributing to **two** enterprise client onboardings.
- Improved recommendation accuracy through feature engineering on student profiles across GPA, test scores, budget and geography preferences.
- Rebuilt **MongoDB** schema from scratch and scripted Python pipeline normalizing **20,000+** data points; scraped and cleaned **2,000+** external records to expand the recommendation corpus.
- Deployed automated discrepancy detection pipeline that caught DB errors daily and triggered developer alerts, replacing a fully manual process.

## Projects

### GPU Scheduler with Multi-Agent RL | Ray RLlib, MAPPO, Kubernetes

Dec 2025 – Present

- GPU nodes negotiate workload allocation via multi-agent RL (MAPPO); benchmarked against Kubernetes baseline.

### Cryptographic Model Fingerprinting | Stable Diffusion XL, Llama 2/3, CleverHans, PyTorch

Aug 2025 – Present

- Embedded cryptographic signatures in model weights to trace AI-generated content origins; robust to compression, attacks and fine-tuning.

### GPU Scheduler with Multi-Agent RL | Stable-Baselines3, scikit-learn, OpenAI Gymnasium

Jan 2026 – Present

- Supervised classifier identifying DQN, PPO or A2C from behavioral observations only - no weights, no logs.

## Open Source

### Streamlit Login/Sign-Up UI | PyPI Library

Aug 2022 – Present

- Open-source auth library for Streamlit apps; **30,000+** downloads with special acknowledgement from the Streamlit team.

### Backlog Manager | VS Code Extension

Jan 2025 – Present

- In-editor task tracking extension with auto-prioritization, one-click code navigation and backlog export.

### Algorithm Visualizer | Python, tkinter, pygame

Nov 2021 – Jan 2022

- Data structures visualization tool with dependency-free executable; research published in **IEEE**.